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Rhodium Catalysis

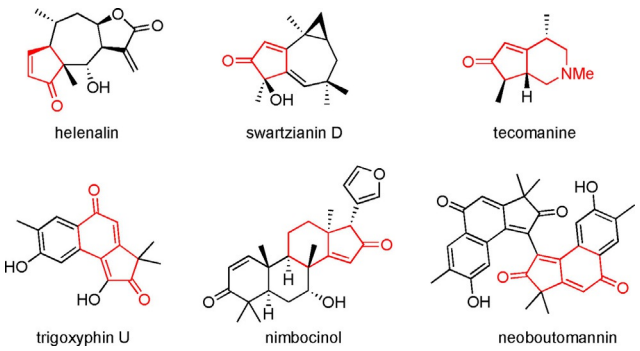
Rhodium-Catalyzed Pauson–Khand-Type Cyclization of 1,5-Allene-Alkynes: A Chirality Transfer Strategy for Optically Active Bicyclic Ketones

Yulin Han,[a] Yichuan Zhao,[b] and Shengming Ma*[a, b]

Abstract: An efficient chirality transfer in the $[RhCl(CO)_2]_2$-catalyzed [2+2+1] cyclization of optically active axially chiral 1,3-disubstituted allenynes with CO to access optically active bicyclopentenone compounds has been developed. The distal C=C bond of allenes reacted with the alknye unit and CO to afford [4.3.0]-bicyclic products with high *ee* values under mild reaction conditions with an excellent selectivity.

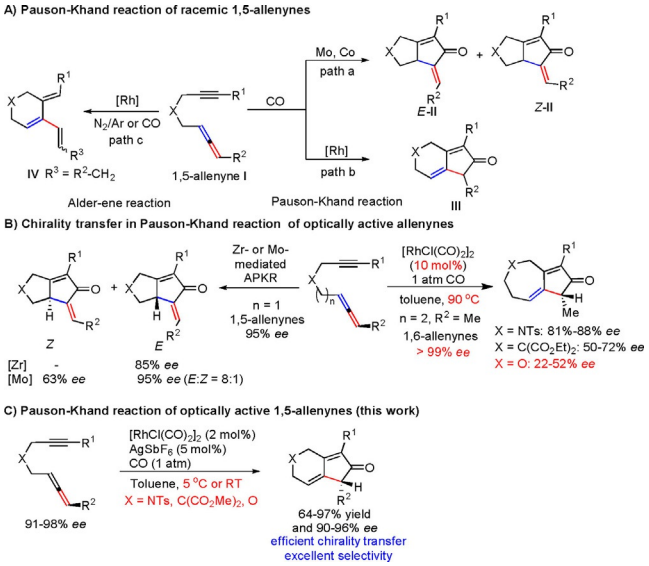
Introduction

Cyclopentenones are present in many important natural products and bioactive compounds (Scheme 1).[1–3] For example, helenalin and its derivatives have anti-cancer and anti-inflammatory effects, showing potential medical applications in the treatment of deregulated and unwanted T-cell-mediated immune responses.[2] In addition, tecomanine has a powerful hypoglycemic activity.[3]



Scheme 1. Some typical bioactive and natural compounds containing a cyclopentenone unit.

The cyclopentenones are also useful building blocks or key intermediates in total synthesis due to the versatility of the enone moiety. The Pauson–Khand reaction between an alkyne and an alkene is one of the most powerful methods to synthesize the cyclopentenone skeleton.[4] When the alkene is replaced with an allene, both C=C bonds of the allene unit may react with the alkyne moiety. Thus, the control of which double bond of the allene moiety reacts in the allenic Pauson–Khand reaction (APKR) of 1,5-allenynes **I** is still a challenge (Scheme 2 A, paths a and b).[4c,5] Notably, in the Co-,[6] Mo-,[7] and Ir-catalyzed[8] APKR, cyclopentenone derivatives **II** were the major products. Brummond and Mukai et al. reported the Rh-catalyzed reaction of allenynes with CO to produce cyclopentenone compounds **III**, in which the distal C=C bond of allenes



Scheme 2. Rhodium-catalyzed allenic Pauson–Khand reactions.

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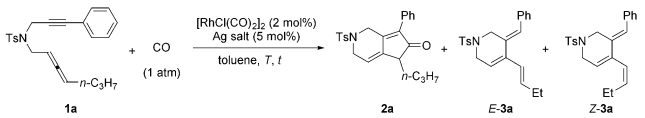
exclusively reacts with the alknye unit and CO (Scheme 2A, path b).[7e–g,9,10] Formal Alder-ene products, cross-conjugated trienes **IV**, were also formed under inert atmosphere as reported by Brummond[11a] and Shibata.[11b] Mukai[11c] and Brummond[7f] also found that the Alder-ene products were obtained as competing products under CO atmosphere through an oxidative cyclometalation/β-H elimination/reductive elimination sequence (Scheme 2A, path c).

On the other hand, in principle, chiral cyclopentenone compounds could be accessed by chirality transfer when the optically active axially chiral allenynes are used. To our knowledge, there are only limited examples on the APKR of chiral allenynes.[7h,12,13] In 2002, Brummond et al. demonstrated a partial transfer of chirality for 1,5-allenynes in the Zr- or Mo-mediated APKR to afford 2-cyclopentenones with the stereogenic center beta to the carbonyl group (Scheme 2B, left). In 2013, the same group reported the Rh^{I}-catalyzed APKR of 1,6-allenynes, which afforded 5,7-bicyclic cyclopentenones with the stereogenic center adjacent to the carbonyl group with a low *ee* value probably due to sensitive α-H (Scheme 2B, right). So far, such reactions of 1,5-allenynes with an acceptable level of chirality transfer have not been developed. Thus, the main challenges for this transformation would be the efficiency of the chirality transfer and the easy racemization of the products due to the sensitive α-H in the bicyclic ketone products. Herein, the Rh-catalyzed APKR of 1,3-disubstituted allenynes to synthesize optically active bicyclopentenones with excellent enantioselectivities under mild reaction conditions is reported (Scheme 2C).

Results and Discussion

We started this study by using racemic 1,3-disubstituted allenyne **1a** as the model substrate. After treatment with $[RhCl(CO)_2]_2$ at 80 °C, the cross-conjugated triene **3a** was formed in 76% yield as the major product (Table 1, entry 1). Interestingly, the desired product **2a**, arising from the reaction between the distal C=C bond of the allene with the alkyne and CO, was formed in 20% yield. There was no improvement in neither the yield of **2a** nor the selectivity when the reaction was conducted at 60 °C (entry 2). When cationic rhodium complex $[Rh(cod)_2]OTf$ was used as the catalyst, the selectivity of APKR and the yield of **2a** were noticeably improved (entry 3). Next, the effect of silver salts was investigated (entries 4–10). The reaction was partially inhibited and yield of **2a** was lower when AgOAc or Ag_3PO_4 were added (entries 8 and 10). The selectivity and the yield of **2a** decreased when AgOTf was used (entry 6), albeit there was a slight improvement when $AgBF_4$ or $AgClO_4$ were used (entries 4 and 7). The yield of **3a** dropped to less than 10% with the addition of $AgSbF_6$ or $AgOCOCF_3$ (entries 5 and 9). The selectivity of **2a** over **3a** was further improved when the reaction was carried out at 10 °C in the presence of $AgSbF_6$ (entry 11). **3a** was not detected when the reaction was performed at 5 °C (entry 12). Thus, the reaction of **1a** using $[RhCl(CO)_2]_2$ (2 mol%) and $AgSbF_6$ (5 mol%) in toluene at 5 °C was chosen as the standard conditions for further study.

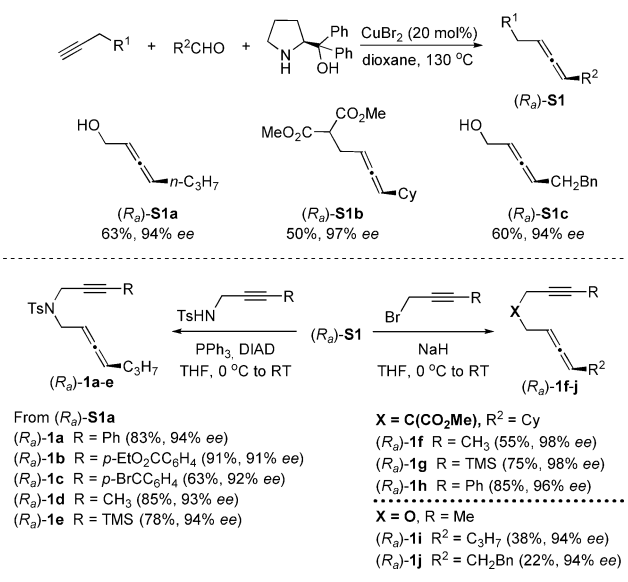
Table 1. Optimization of reaction parameters for the rhodium-catalyzed Pauson–Khand cyclization of racemic 1,5-allene-alkyne **1a**.



Entry	[Ag]	*T* [°C]	*T* [h]	Yield of **2a** [%][b]	Yield of **3a** (*E*/*Z*) [%][b]	Recovery of **1a** [%][b]
1[a]	–	80	4	19 (20)[c]	74 (76)[c] (3:1)	–
2[a]	–	60	4	18	77 (3.3:1)	–
3[a,d]	–	RT	13	53	37 (2.4:1)	–
4[e]	$AgBF_4$	RT	24	60	22 (3.4:1)	3
5[e]	$AgSbF_6$	RT	4	68	10 (4:1)	–
6[e]	AgOTf	RT	4	44	36 (2.6:1)	–
7[e]	$AgClO_4$	RT	2.5	61	27 (2:1)	–
8[e]	AgOAc	RT	21	4	7 (2.5:1)	65
9[e]	$AgOCOCF_3$	RT	21	66	8 (1.7 :1)	–
10[e]	Ag_3PO_4	RT	24	13	59 (3.5:1)	28
11[e]	$AgSbF_6$	10	7	92	11 (2.7:1)	–
12[e]	$AgSbF_6$	5	11	89	–	–

[a] The reaction was performed with **1a** (0.2 mmol), catalyst (2 mol%) in 2 mL of toluene. [b] Determined by ^1H NMR analysis with mesitylene as the internal standard. [c] Isolated yield. [d] $[Rh(cod)_2]OTf$ (2 mol%) was used. [e] The reaction was carried out with **1a** (0.2 mmol), cationic catalyst (2 mol%, which was generated in situ after stirring a suspension of $[RhCl(CO)_2]_2$ and Ag salt for 10 mins) in 2 mL of toluene.

(R_a)-**1** were prepared by substitution reaction of the optically active allenols (R_a)-**S1a** and (R_a)-**S1c** or allenylmalonate (R_a)-**S1b**, which were prepared by a $CuBr_2$-catalyzed EATA (enantioselective allenation of terminal alkynes) reaction[14] (Scheme 3). Based on the results shown in Table 1, the reaction of optically active allenyne (R_a)-**1a** (93% *ee*) was conducted under the standard conditions. To our delight, (*R*)-**2a** was formed with 89% *ee* and 80% yield after 11 h (Table 2, entry 1). (*R*)-**2a** was formed in 79% yield with a complete chirality transfer when the reaction time was shortened from 11 to 9 h, indicating an in situ racemization caused by the sensitive α-H (entry 2). The abso-

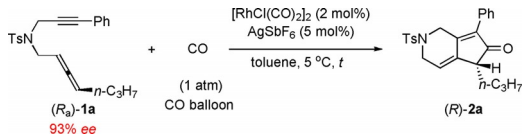


Scheme 3. Synthesis of (R_a)-**1**.

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Table 2. Pauson–Khand-type cyclization of optically active allene-alkyne (R_a)-**1a**.[a]



Entry	*t* [h]	(*R*)-**2a** Yield [%][b]	*ee* [%]
1	11	80	89
2	9	79	94

[a] The reaction was performed with (R_a)-**1a** (0.2 mmol), $[RhCl(CO)_2]_2$ (2 mol%) and $AgSbF_6$ (5 mol%) in 2 mL of toluene. [b] Isolated yield.

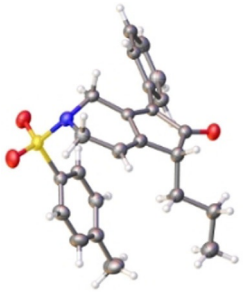
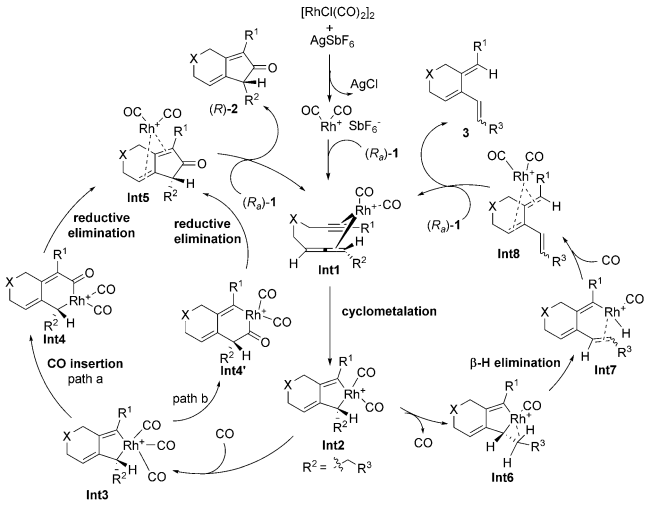


Figure 1. ORTEP representation of (*R*)-**2a**.

lute configuration of (*R*)-**2a** was established by single-crystal X-ray diffraction analysis (Figure 1).[15]

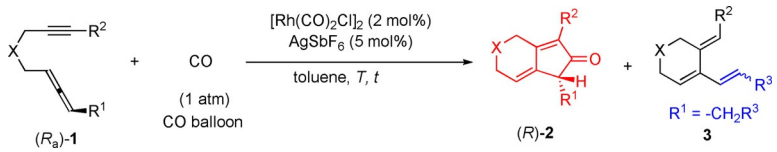
Next, the scope of the reaction was investigated. A variety of substrates containing NTs as the linker afforded the products with yields of about 90% and an excellent chirality transfer, whereas the corresponding triene products **3** were formed in <1% yield (Table 3, entries 1–5). Remarkably, excellent results were obtained for the allenynes bearing an internal alkyne substituted with Ph, CH_3, as well as other synthetically useful groups such as CO_2Et, TMS and Br. The substrates containing malonate as the linker could also afford the desired products with excellent chirality transfer when the reaction was carried out at room temperature (entries 6–8). Although the reaction of (R_a)-**1h** afforded 8% yield of cross-conjugated triene **3h**, the efficiency of chirality transfer for the formation of **2h** is still excellent (entry 8, Table 3). Finally, the oxygen atom could also be used as the linker for this reaction, affording the corresponding products (*R*)-**2i**–**2j** with high *ee* values (entries 9–10).

Based on the X-ray diffraction analysis of (*R*)-**2a** and previous reports,[13,16] a mechanism of the Rh-catalyzed APKR is proposed (Scheme 4). Firstly, $[RhCl(CO)_2]_2$ reacts with $AgSbF_6$ to generate cationic catalyst $[Rh(CO)_2]^+SbF_6^-$. Highly selective coordination of the distal C=C bond and the C≡C bond in alle-



Scheme 4. Proposed mechanism for the rhodium-catalyzed allenic Pauson–Khand reaction of 1,3-disubstituted allenynes.

Table 3. Scope of the rhodium-catalyzed Pauson–Khand reaction of optically active 1,3-disubstituted allenynes.[a]



Entry	X	(R_a)-**1** R^1	R^2	*ee* [%]	*T* [°C]	*t* [h]	(*R*)-**2** Yield [%]	*ee* [%]	Yield of **3** [%][b]
1	NTs	*n*-C_3H_7	Ph (**1a**)	94	5	9	94 (**2a**)	94	<1
2	NTs	*n*-C_3H_7	*p*-$EtO_2CC_6H_4$ (**1b**)	91	RT	1	88 (**2b**)	92	3
3	NTs	*n*-C_3H_7	*p*-BrC_6H_4 (**1c**)	92	RT	1	93 (**2c**)	90	<1
4	NTs	*n*-C_3H_7	CH_3 (**1d**)	93	5	9	91 (**2d**)	92	<1
5	NTs	*n*-C_3H_7	TMS (**1e**)	94	5	3.5	97 (**2e**)	94	<1
6[c]	$C(CO_2Me)_2$	Cy	CH_3 (**1f**)	98	RT	1.5	84 (**2f**)	96	<1
7	$C(CO_2Me)_2$	Cy	TMS (**1g**)	97	RT	2.5	96 (**2g**)	93	<1
8[c]	$C(CO_2Me)_2$	Cy	Ph (**1h**)	96	RT	4	64 (**2h**)	94	8
9	O	*n*-C_3H_7	CH_3 (**1i**)	94	RT	2	81 (**2i**)	92	<1
10	O	CH_2Bn	CH_3 (**1j**)	94	RT	4	85 (**2j**)	91	<1

[a] The reaction was carried out with (R_a)-**1** (0.5 mmol), $[RhCl(CO)_2]_2$ (2 mol%), and $AgSbF_6$ (5 mol%) in 5 mL of toluene at 5 °C. [b] Isolated yield. [c] The reaction was performed on 0.2 mmol scale.

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nyne (R_a)-**1** with the cationic catalyst affords **Int1**. Subsequent cyclometallation leads to the formation of Rh^{III} intermediate **Int2**. Coordination of CO to the Rh^{III} leads to the formation of highly reactive transient intermediate **Int3**, which would immediately undergo CO insertion to afford **Int4** and/or **Int4′**. After reductive elimination **Int5** would be afforded. The ligand exchange between **Int5** and (R_a)-**1** provides the cyclopentenone product (*R*)-**2** and regenerates **Int1** to complete the catalytic cycle. The β-H elimination of **Int2** would afford **Int7** via intermediate **Int6**. Subsequent reductive elimination and ligand exchange provides cross-conjugated triene **3** as the byproduct.

Conclusions

A $[RhCl(CO)_2]_2/AgSbF_6$ co-catalyzed APKR of optically active axially chiral 1,3-disubstituted allenynes to prepare chiral cyclopentenone compounds with an excellent efficiency of chirality transfer has been developed. Alkyl, aryl as well as other common functional groups, such as silyl, ester and halogens, are tolerated. The mild reaction conditions, broad scope, and excellent selectivity make this reaction an efficient method to build the chiral cyclopentenone skeleton.

Experimental Section

To a flame-dried Schlenk tube were added $[RhCl(CO)_2]_2$ (3.8 mg, 0.01 mmol) and $AgSbF_6$ (8.8 mg, 0.025 mmol) inside a glove box. After being evacuated and refilled with CO (gas) for three times using a balloon outside the glove box, to the Schlenk tube were added 3.0 mL of toluene. The Schlenk tube was then placed in an ethanol bath pre-cooled to 5 °C. After being stirred for 10 min, a solution of (R_a)-**1a** (189.7 mg, 0.5 mmol, 94% *ee*) in 2.0 mL of toluene was added dropwise using a syringe to the mixture over 1 min. The resulting mixture was stirred at 5 °C for 9 h until the reaction was complete as monitored by TLC. The crude reaction mixture was filtrated through a pad of silica gel eluted with diethyl ether (30 mL). After concentration in vacuo, the residue was purified by column chromatography on silica gel (petroleum ether/ethyl acetate = 5:1) to afford (*R*)-**2a** (190.9 mg, 94%) as a solid; m.p. 159–160 °C (petroleum ether/ethyl acetate); $[\alpha]_D^{30}$ +10.56 (c = 1.000 in $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.57 (d, J = 8.4 Hz, 2H, Ar-H), 7.46 (t, J = 7.4 Hz, 2H, Ar-H), 7.42–7.37 (m, 1H, Ar-H), 7.36–7.32 (m, 2H, Ar-H), 7.25 (d, J = 8.8 Hz, 2H, Ar-H), 5.86 (t, J = 3.8 Hz, 1H, HC=), 4.60 (d, J = 17.2 Hz, 1H, one proton of NCH_2), 4.45 (d, J = 17.2 Hz, 1H, one proton of NCH_2), 4.13 (dd, J = 18.2 Hz, 3.8 Hz, 1H, one proton of NCH_2), 4.05 (dd, J = 18.4 Hz, 3.2 Hz, 1H, one proton of NCH_2), 2.72 (t, J = 6.0 Hz, 1H, CH), 2.40 (s, 3H, CH_3), 1.62–1.54 (m, 1H, one proton of CH_2), 1.53–1.42 (m, 1H, one proton of CH_2), 1.29–1.17 (m, 2H, CH_2), 0.86 ppm (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 204.7, 154.8, 144.0, 138.9, 136.5, 134.2, 130.0, 129.7, 128.69, 128.66, 128.63, 127.5, 119.1, 46.8, 44.7, 44.0, 32.0, 21.5, 19.2, 14.1 ppm; MS (ESI) m/z: 430 ($[M+Na]^+$), 408 ($[M+H]^+$); IR (neat): $\tilde{\nu}$ = 3055, 2960, 2932, 2875, 1691, 1594, 1490, 1441, 1340, 1306, 1231, 1153, 1090 cm^{-1}; elemental analysis calcd (%) for $C_{24}H_{25}NO_3S$: C 70.73, H 6.18, N 3.44; found: C 70.54, H 6.24, N 3.32.

HPLC (Chiralcel AD-H column, *n*-hexane/*i*PrOH = 95:5; flow rate: 1.2 mL min^{-1}, λ = 214 nm): t_R = 10.87 min (minor); t_R = 14.64 min (major); 94% *ee*.

Acknowledgements

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Conflict of interest

The authors declare no conflict of interest.

Keywords: allenynes · allenic Pauson–Khand reaction · bicyclic ketones · chirality · rhodium

[1] S. P. Simeonov, P. M. Nunes, K. Guerra, V. B. Kurteva, C. A. M. Afonso, *Chem. Rev.* **2016**, *116*, 5744.
[2] C. Berges, D. Fuchs, G. Opelz, V. Daniel, C. Naujokat, *Mol. Immunol.* **2009**, *46*, 2892.
[3] D. A. Ockey, M. A. Lewis, N. E. Schore, *Tetrahedron* **2003**, *59*, 5377.
[4] For a recent book on Pauson–Khand-type reaction, see: a) *The Pauson–Khand reaction: scope, variations and applications* (Ed.: R. R. Torres), Wiley, Chichester, **2012**. For selected reviews, see: b) H.-W. Lee, F.-Y. Kwong, *Eur. J. Org. Chem.* **2010**, 789; c) T. Shibata, *Adv. Synth. Catal.* **2006**, *348*, 2328.
[5] a) B. Alcaide, P. Almendros, *Eur. J. Org. Chem.* **2004**, 3377; b) S. Kitagaki, F. Inagaki, C. Mukai, *Chem. Soc. Rev.* **2014**, *43*, 2956.
[6] For selected examples on Co-catalyzed intramolecular allenic Pauson–Khand reaction, see: B. L. Pagenkopf, D. B. Belanger, D. J. R. O'Mahony, T. Livinghouse, *Synthesis* **2000**, 1009.
[7] For selected examples on Mo-catalyzed intramolecular allenic Pauson–Khand reaction, see: a) J. L. Kent, H. Wan, K. M. Brummond, *Tetrahedron Lett.* **1995**, *36*, 2407; b) K. M. Brummond, H. Wan, *Tetrahedron Lett.* **1998**, *39*, 931; c) K. M. Brummond, H. Wan, J. L. Kent, *J. Org. Chem.* **1998**, *63*, 6535; d) K. M. Brummond, P. C. Sill, B. Reckards, S. J. Geib, *Tetrahedron Lett.* **2002**, *43*, 3735; e) K. M. Brummond, H. Chen, K. D. Fisher, A. D. Kerekes, B. Rickards, P. C. Sill, S. J. Geib, *Org. Lett.* **2002**, *4*, 1931; f) K. M. Brummond, D. Chen, *Org. Lett.* **2008**, *10*, 705; g) K. M. Brummond, B. Mitasev, *Org. Lett.* **2004**, *6*, 2245; h) K. M. Brummond, A. D. Kerekes, H. Wan, *J. Org. Chem.* **2002**, *67*, 5156.
[8] T. Shibata, S. Kadowaki, M. Hirose, K. Takagi, *Synlett* **2003**, 0573.
[9] For selected examples on Rh-catalyzed intramolecular allenic Pauson–Khand reaction, see: a) T. Kobayashi, Y. Koga, K. Narasaka, *J. Organomet. Chem.* **2001**, *624*, 73; b) C. Mukai, I. Nomura, S. Kitagaki, *J. Org. Chem.* **2003**, *68*, 1376; c) F. Inagaki, T. Kawamura, C. Mukai, *Tetrahedron* **2007**, *63*, 5154; d) K. M. Brummond, M. M. Davis, C. Huang, *J. Org. Chem.* **2009**, *74*, 8314; e) C. Mukai, F. Inagaki, T. Yoshida, K. Yoshitani, Y. Hara, S. Kitagaki, *J. Org. Chem.* **2005**, *70*, 7159.
[10] For selected examples on other metal catalyzed intramolecular allenic Pauson–Khand reaction to afford III, as major product, see: a) M. Ahmar, C. Locatelli, D. Colombier, B. Cazes, *Tetrahedron Lett.* **1997**, *38*, 5281; b) H. Xiong, R. P. Hsung, L.-L. Wei, C. R. Berry, J. A. Mulder, B. Stockwell, *Org. Lett.* **2000**, *2*, 2869.
[11] a) K. M. Brummond, H. Chen, P. C. Sill, L. You, *J. Am. Chem. Soc.* **2002**, *124*, 15186; b) T. Shibata, Y. Takesue, S. Kadowaki, K. Takagi, *Synlett* **2003**, 0268; c) C. Mukai, F. Inagaki, T. Yoshida, S. Kitagaki, *Tetrahedron Lett.* **2004**, *45*, 4117.
[12] For the report on Rh-catalyzed APKR of allenynes with a low efficiency of chirality transfer, see: 1,6-allenynes: F. Grillet, K. M. Brummond, *J. Org. Chem.* **2013**, *78*, 3737.
[13] L. C. Burrows, L. T. Jesikiewicz, G. Lu, S. J. Geib, P. Liu, K. M. Brummond, *J. Am. Chem. Soc.* **2017**, *139*, 15022.
[14] X. Huang, T. Cao, Y. Han, X. Jiang, W. Lin, J. Zhang, S. Ma, *Chem. Commun.* **2015**, *51*, 6956.
[15] Crystal data for compound (R_a)-**2a**: $C_{24}H_{25}NO_3S$, MW = 407.51, orthorhombic, space group P 21 21 21, final *R indices* [$I > 2\sigma(I)$], $R1$ = 0.0278,

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*wR*2 = 0.0733, *R* indices (all data) *R*1 = 0.0283, *wR*2 = 0.0739, *a* = 7.9161(1) Å, *b* = 8.3633(1) Å, *c* = 31.9114(6) Å, $\alpha = 90^\circ$, $\beta = 90^\circ$, $\gamma = 90^\circ$, *V* = 2112.69(5) Å^3, *T* = 173 K, *Z* = 4, reflections collected/unique 26511/4003 (R_{int} = 0.0336), number of observations [$> 2\sigma(I)$] 4003, parameters: 264. CCDC 1574968 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre.

[16] A. S. Bayden, K. M. Brummond, K. D. Jordan, *Organometallics* **2006**, *25*, 5204.

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